

SQL styling tool

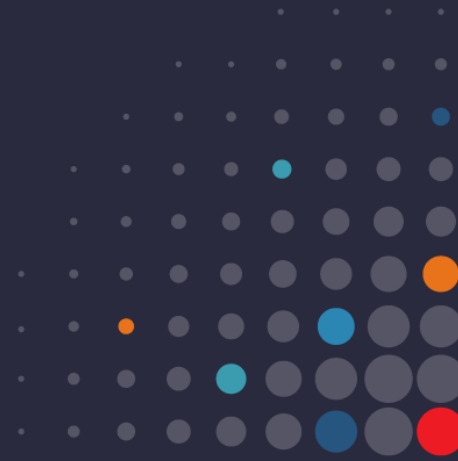
By Daniel Contractor

February 2025

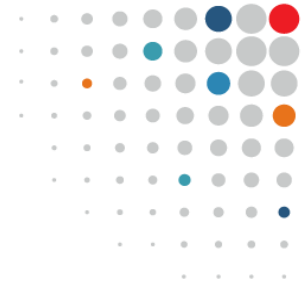


Social Investment Agency
Toi Hau Tāngata

Te Kāwanatanga o Aotearoa
New Zealand Government



Overview of tool



Need

- Analytics team produces SQL definitions with sharing potential (Github library)
 - Not well-presented enough for reuse
 - Team has very limited time to tidy code for publication

Purpose

- Makes code easier to re-use and share
 - Improves presentation of code
 - No existing tool in IDI

Application

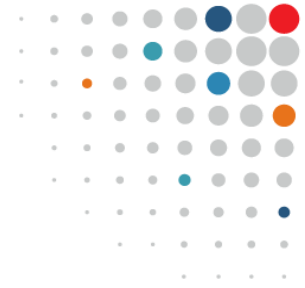
- Styled 200+ files (in 15 seconds batch process)

Key features

Standardization approach

- Style inspired ‘commas before’ by *Redgate*
 - Places commas at the start of lines
 - Performs multi-layer indentation
 - Capitalizes keywords (e.g. SELECT, FROM..)
 - Leaves functions & comments untouched

Example use: indentation



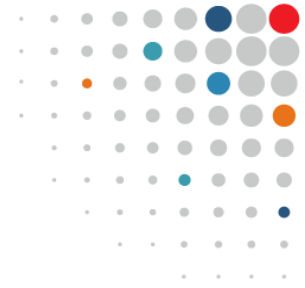
Before

```
WITH multi_to_add AS (  
  SELECT snz_uid  
  FROM [IDI_Sandpit].[DL-MAA20XX-YY].[tmp_bipolar_multi] AS m  
  WHERE EXISTS (  
    SELECT 1  
    FROM [IDI_Sandpit].[DL-MAA20XX-YY].[tmp_bipolar_solo] AS s  
  )  
)
```

After

```
WITH multi_to_add AS (  
  SELECT snz_uid  
  FROM [IDI_Sandpit].[DL-MAA20XX-YY].[tmp_bipolar_multi] AS m  
  WHERE EXISTS (  
    SELECT 1  
    FROM [IDI_Sandpit].[DL-MAA20XX-YY].[tmp_bipolar_solo] AS s  
  )  
)
```

Example use: lists



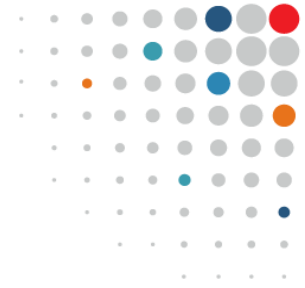
Before

```
SELECT [snz_uid], [moh_mhd_activity_start_date],  
       [moh_mhd_activity_end_date], [moh_mhd_activity_type_code],  
       [moh_mhd_activity_status_code], [moh_mhd_activity_unit_type_text]
```

After

```
SELECT [snz_uid]  
       , [moh_mhd_activity_start_date]  
       , [moh_mhd_activity_end_date]  
       , [moh_mhd_activity_type_code]  
       , [moh_mhd_activity_status_code]  
       , [moh_mhd_activity_unit_type_text]
```

Example use: capitalisation



Before

```
where ([CLINICAL_ID] >= 10 and substring([CODE],1,3) in ('F30','F31'))  
/* Select staff where recommend = 7 instead of >= 7 */
```

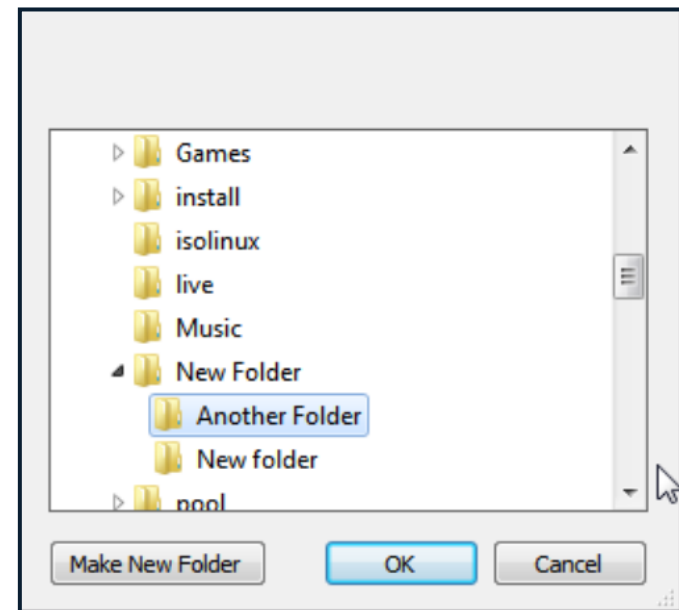
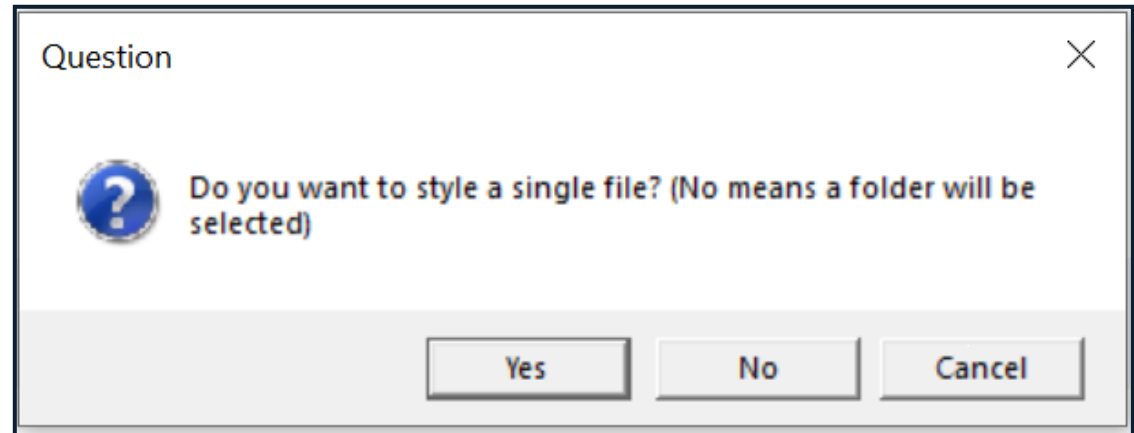
After

```
WHERE ([CLINICAL_ID] >= 10 AND SUBSTRING([CODE],1,3) IN ('F30','F31'))  
/* Select staff where recommend = 7 instead of >= 7 */
```

Tool components

How to use

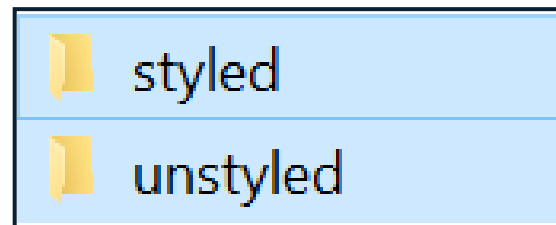
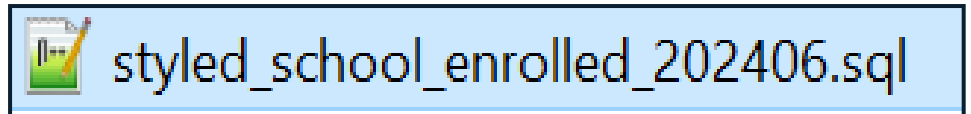
- Run main script
- Select input file(s)
- Select output file(s)
- Debug



Output structure

How styled files are saved

- Output files
 - Prefixed with 'styled_'
- Output folders
 - Unstyled folder
 - Styled folder



Thank you!

info@sia.govt.nz



Social Investment Agency
Toi Hau Tāngata

Te Kāwanatanga o Aotearoa
New Zealand Government

